

LOAN DEFAULT RISK PREDICTION USING MACHINE LEARNING

Business Analytics / Machine Learning Project

1. Executive Summary

Loan default is a major risk for banks and financial institutions. This project develops a machine-learning decision-support prototype using the UCI Statlog German Credit dataset. The workflow includes data preparation, exploratory analysis, model comparison, evaluation, model saving and an interactive Streamlit application.

2. Business Problem

Lenders need to estimate credit risk before approving loans. The project predicts whether an application will be classified as a good or bad credit risk. The model is intended to support, not replace, formal underwriting.

3. Objectives

The objectives are to select a reliable public dataset, prepare the data, perform EDA, compare classification models, evaluate performance, save the best model, deploy an interactive application and provide business recommendations.

4. Dataset

The project uses the UCI Statlog (German Credit Data), dataset ID 144. UCI reports 1,000 instances and 20 features. The data contain categorical and integer variables describing credit applications. Source: UCI Machine Learning Repository, <https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data>

5. Data Preparation

The target is converted into binary form. Numerical variables are imputed and standardized. Categorical variables are imputed and one-hot encoded. An 80:20 stratified train-test split is used. Preprocessing is included in the model pipeline so training and prediction use consistent transformations.

6. Exploratory Data Analysis

The notebook examines class distribution, numerical summaries, loan duration, credit amount, age and applicant characteristics. The purpose is to identify patterns associated with higher credit risk and to support the business interpretation.

7. Machine Learning

Logistic Regression is used as an interpretable baseline and Random Forest as a non-linear ensemble model. Both are evaluated with Accuracy, Precision, Recall, F1-score and ROC-AUC. The training script selects the best model using ROC-AUC, with recall as a secondary consideration.

8. Model Evaluation

The exact numerical results are generated by `train_model.py` and saved to `models/model_metrics.json`. This report intentionally does not invent performance figures. After running the training script, the actual metrics can be inserted into this section or referenced from the generated JSON file.

9. Business Recommendations

High-risk applications should receive enhanced underwriting, verification of income and existing obligations, and careful review of repayment capacity. Medium-risk cases can receive additional affordability checks. Lower-risk cases can proceed through normal underwriting subject to policy.

10. Streamlit Application

The application accepts 20 applicant attributes, calculates the model probability, displays a risk category and provides a business recommendation. It includes a disclaimer that the model is an academic decision-support prototype.

11. Deployment

The repository is structured for GitHub and Streamlit Community Cloud. Install dependencies, run `train_model.py` to create the model, then run `streamlit run app.py`. The same `app.py` can be selected as the Streamlit deployment entry point.

12. Limitations

The dataset contains 1,000 historical observations and represents an older credit environment. Current lending conditions may differ. The model should therefore not be treated as a production credit approval system without validation, monitoring, fairness testing, explainability and regulatory review.

13. Future Scope

Future work can include gradient boosting, hyperparameter tuning, cross-validation, calibration, cost-sensitive threshold optimization, SHAP explanations, fairness analysis, model monitoring and integration with a loan-origination platform.

14. Conclusion

The project demonstrates the full business analytics ML lifecycle: Business Problem → Dataset → Data Preparation → ML Model → Evaluation → Prediction → Business Recommendation → Streamlit App → GitHub → Deployment.

Project Structure

File	Purpose
app.py	Interactive Streamlit application
train_model.py	Dataset loading, preprocessing, training and evaluation
notebooks/Loan_Default_Analysis.ipynb	EDA and model development notebook
models/loan_default_model.pkl	Generated trained model after running train_model.py
models/model_metrics.json	Generated model-performance metrics
requirements.txt	Python dependencies
README.md	Project documentation

Viva Preparation

- Why is loan default a classification problem?
- Why were Logistic Regression and Random Forest selected?
- Why are recall and ROC-AUC important in credit-risk modelling?
- What is a false negative in this business context?
- Why should a model not be the only basis for a real loan decision?
- What are the main limitations of the dataset?
- How can the model be improved in future?